

Homework 01

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Problem 1. *Let E and F be events. Using the axioms of probability, prove that*

$$P(E \cup F) = P(E) + P(F) - P(EF).$$

E is carved by F into two disjoint pieces:

$$E = EF \cup EF^c \quad (\text{disjoint}),$$

so by axiom 3,

$$P(E) = P(EF) + P(EF^c) \iff P(EF^c) = P(E) - P(EF).$$

Similarly, $P(FE^c) = P(F) - P(EF)$.

Therefore, since $E \cup F = EF \cup EF^c \cup FE^c$ is a union of pairwise disjoint sets,

$$\begin{aligned} P(E \cup F) &= P(EF \cup EF^c \cup FE^c) \\ &= P(EF) + P(EF^c) + P(FE^c) \\ &= P(EF) + (P(E) - P(EF)) + (P(F) - P(EF)) \\ &= P(E) + P(F) - P(EF). \end{aligned}$$

Problem 8. *If $P(E) = 0.9$ and $P(F) = 0.8$, show that $P(EF) \geq 0.7$. In general, show that*

$$P(EF) \geq P(E) + P(F) - 1.$$

This is known as Bonferroni's inequality.

First we record the bound the argument rests on.

Lemma. *For any event A , $P(A) \leq 1$.*

Since $\Omega = A \cup A^c$ is a union of disjoint sets,

$$P(\Omega) = 1 = P(A \cup A^c) = P(A) + P(A^c) \quad (\text{disjoint}),$$

so

$$P(A) + P(A^c) = 1 \iff P(A) = 1 - P(A^c) \implies P(A) \leq 1,$$

since $P(A^c) \geq 0$ by the first axiom.

Now prove $P(EF) \geq P(E) + P(F) - 1$. Recall, using Problem 1,

$$1 \geq P(E \cup F) = P(E) + P(F) - P(EF),$$

and therefore

$$1 + P(EF) \geq P(E) + P(F) \iff P(EF) \geq P(E) + P(F) - 1.$$

Therefore, if $P(E) = 0.9$ and $P(F) = 0.8$,

$$P(EF) \geq 0.9 + 0.8 - 1 \geq 0.7.$$

Problem 10. Show that

$$P\left(\bigcup_{i=1}^n E_i\right) \leq \sum_{i=1}^n P(E_i).$$

This is known as Boole's inequality.

Hint: Either use Equation (1.2) and mathematical induction, or else show that $\bigcup_{i=1}^n E_i = \bigcup_{i=1}^n F_i$, where $F_1 = E_1$, $F_i = E_i \cap \bigcap_{j=1}^{i-1} E_j^c$, and use property (iii) of a probability.

Problem 12. Let E and F be mutually exclusive events in the sample space of an experiment. Suppose that the experiment is repeated until either event E or event F occurs. What does the sample space of this new super experiment look like? Show that the probability that event E occurs before event F is $P(E)/[P(E) + P(F)]$.

Hint: Argue that the probability that the original experiment is performed n times and E appears on the n th time is $P(E) \times (1 - p)^{n-1}$, $n = 1, 2, \dots$, where $p = P(E) + P(F)$. Add these probabilities to get the desired answer.

Problem 39. Stores A , B , and C have 50, 75, and 100 employees and, respectively, 50, 60, and 70 percent of these are women. Resignations are equally likely among all employees, regardless of sex. One employee resigns and this is a woman. What is the probability that she works in store C ?

Problem 41. In a certain species of rats, black dominates over brown. Suppose that a black rat with two black parents has a brown sibling.

- (a) What is the probability that this rat is a pure black rat (as opposed to being a hybrid with one black and one brown gene)?
- (b) Suppose that when the black rat is mated with a brown rat, all five of their offspring are black. Now, what is the probability that the rat is a pure black rat?